LOCALIZING A MORPHISM OF SITES WHEN THE FUNCTOR COMMUTES WITH FINITE LIMITS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The proposition producing a morphism of sites from a functor. Same chapter, Proposition proposition-get-morphism. Produces the two morphisms of sites appearing in the diagram.
- The localization of a site when the site has products. Same chapter, Lemma lemma-localize-given-products. Supplies the description of the horizontal arrows of the diagram.
- Composition of morphisms of sites. Same chapter, Definition definition-composition-morphisms-sites. Supplies the commutativity of the diagram of categories and functors.

Source reference. Original lemma 03CE: sources/stacks-sites/pinned-original-source.tex, source lines 6629–6629. Navigation pointer invoked at proof line 6629; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03CF: sources/stacks-sites/pinned-original-source.tex, source lines 6601–6601. Navigation pointer invoked at proof line 6601; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original proposition 00X6: sources/stacks-sites/pinned-original-source.tex, source lines 6638–6638. Navigation pointer invoked at proof line 6638; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original definition 03CC: sources/stacks-sites/pinned-original-source.tex, source lines 6645–6645. Navigation pointer invoked at proof line 6645; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$, $\mathcal{D}$ be sites. Let $u : \mathcal{D} \to \mathcal{C}$ be a functor. Let $V \in \mathrm{Ob}(\mathcal{D})$. Set $U = u(V)$. Assume that*

(1) *$\mathcal{C}$ and $\mathcal{D}$ have all finite limits,*
(2) *u is continuous, and*
(3) *u commutes with finite limits.*

There exists a commutative diagram of morphisms of sites

$$\begin{array}{ccc} \mathcal{C}/U & \xrightarrow[j_U]{} & \mathcal{C} \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ \mathcal{D}/V & \xrightarrow{j_V} & \mathcal{D} \end{array}$$

where the right vertical arrow corresponds to u, the left vertical arrow corresponds to the functor $u' : \mathcal{D}/V \to \mathcal{C}/U$, $V'/V \mapsto u(V')/u(V)$ and the horizontal arrows correspond to the functors $\mathcal{C} \to \mathcal{C}/U$, $X \mapsto X \times U$ and $\mathcal{D} \to \mathcal{D}/V$, $Y \mapsto Y \times V$ as in Lemma 03CE. Moreover, the associated diagram of morphisms of topoi is equal to the diagram of Lemma 03CF. In particular we have $f'_* j_U^{-1} = j_V^{-1} f_*$.

Proof. Note that u satisfies the assumptions of Proposition 00X6 and hence induces a morphism of sites $f : \mathcal{C} \to \mathcal{D}$ by that proposition. It is clear that u induces a functor u' as indicated. It is clear that this functor also satisfies the assumptions of Proposition 00X6. Hence we get a morphism of sites $f' : \mathcal{C}/U \to \mathcal{D}/V$. The diagram commutes by our definition of composition of morphisms of sites (see Definition 03CC) and because

$$u(Y \times V) = u(Y) \times u(V) = u(Y) \times U$$

which shows that the diagram of categories and functors opposite to the diagram of the lemma commutes. □